Dong-Kyum Kim

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Research Interests

My research combines AI, physics, and neuroscience to uncover how large language models learn, store, and revise knowledge. Using representation geometry and causal interventions, I study memory, model editing, and machine unlearning, seeking simple mechanisms behind complex behavior and testable hypotheses inspired by biological memory. I aim to extend this approach to continual reinforcement learning by identifying how rewards preserve existing capabilities and selectively consolidate experience, with the goal of building safer, more reliable, and more efficient AI systems.

Education

Korea Advanced Institute of Science and Technology (KAIST) 2016–2022
Ph.D. in Physics — Advisor: Prof. Hawoong Jeong
Dissertation: *Nonequilibrium Statistical Physics Study using Deep Learning*

Seoul National University (SNU) 2011–2015
B.S. in Physics, Minor in Computer Science & Engineering

Research Experience

Postdoctoral Researcher, Max Planck Institute for Security and Privacy (MPI-SP) — Oct. 2024–Present

- Hosted by Prof. Meeyoung Cha (Scientific Director)
- Data Science for Humanity Group

Postdoctoral Researcher, Institute for Basic Science (IBS) — Mar. 2022–Sep. 2024

- Hosted by Prof. Meeyoung Cha (Chief Investigator)
- Data Science Group, Center for Mathematical and Computational Science

Data Science Intern, Samsung Electronics — Sep.–Dec. 2017

- Built distributed ML systems with Daniel Kim (Senior Data Scientist): multi-GPU anomaly image classification (Keras & Spark) and an Elasticsearch-based image search framework

Selected Publications

† denotes equal contribution.

[S1] Jea Kwon†, **Dong-Kyum Kim**†, Jiwon Kim†, Yonghyun Kim, Woong Kook, and Meeyoung Cha. "AI Engram: In Search of Memory Traces in Artificial Intelligence." *ICML* 2026 **(Oral)**.

[S2] **Dong-Kyum Kim**, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. "Bilinear representation mitigates reversal curse and enables consistent model editing." *ICLR* 2026.

[S3] Gwangsu Kim, **Dong-Kyum Kim**, and Hawoong Jeong. "Spontaneous emergence of rudimentary music detectors in deep neural networks." *Nature Communications* 2024.

[S4] **Dong-Kyum Kim**†, Jea Kwon†, Meeyoung Cha, and C. Justin Lee. "Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity." *NeurIPS* 2023.

[S5] **Dong-Kyum Kim**†, Youngkyoung Bae†, Sangyun Lee, and Hawoong Jeong. "Learning Entropy Production via Neural Networks." *Phys. Rev. Lett.* **125**, 140604 (2020).

Teaching Experience

Computational Physics Course, KAIST (Daejeon, Korea) — May 1, 2023
"Deep learning applications: Nonequilibrium statistical physics study using AI"

Computational Physics Course, KAIST (Daejeon, Korea) — Apr. 24, 2023
"Deep Learning Introduction"

SNU Physics and AI Winter School (Seoul, Korea) — Feb. 24, 2022
"Exploring Irreversibility via Machine Learning"

General Physics II, KAIST (Korea) Fall 2016; Spring 2017
Teaching Assistant

Talks

Spark(l)ing Science, MPI-SP (MB/1 Seminar Room + Zoom) Jul. 24, 2026
"AI research using agents and notebooks"

CREx #23 (Online) Feb. 24, 2026
"Bilinear relational structure fixes reversal curse and enables consistent model editing"

IBS Winter School on AI-Boosted Basic Science (Daejeon, Korea) Dec. 13, 2022
"Resemblances between Transformer's Nonlinearity and NMDA Receptor Dynamics"

KIAS CAINS Summer Workshop (Jeju, Korea) Sep. 2, 2022
"Working and reference memory in transformers on a navigation task"

KIAS Nonequilibrium Statistical Physics of Complex Systems (Seoul, Korea) Jul. 25, 2022
"Deep reinforcement learning for optimal mechanism in active Brownian particles"

KIAS AI and Natural Sciences Seminar (Seoul, Korea) Apr. 27, 2022
"Nonequilibrium Statistical Physics Study Using Deep Learning"

APCTP Workshop for Physics and Machine Learning (Jeju, Korea) Nov. 26, 2021
"Exploring optimal mechanisms in active Brownian particles via deep reinforcement learning"

Seoul National University Statistical Physics Seminar (Online) Feb. 1, 2021
"Methods of estimating entropy production"

Korean Physical Society Fall Meeting (Online) Nov. 6, 2020
"Deep reinforcement learning for feedback-controlled flashing ratchets"

NetSci2020 (Online) Sep. 22, 2020
"Discovering wiring patterns of neural networks via backboning"

Korean Physical Society Spring Meeting (Online) Jul. 13, 2020
"Neural estimator for entropy production"

Quantifying Success Satellite at NetSci2019 (Burlington, Vermont, USA) May 27, 2019
"Quantifying Individual Reputation in Large-scale Historical Documents"

Publications

† denotes equal contribution.

[1] Jea Kwon†, **Dong-Kyum Kim**†, Jiwon Kim†, Yonghyun Kim, Woong Kook, and Meeyoung Cha. "AI Engram: In Search of Memory Traces in Artificial Intelligence." *ICML* 2026 **(Oral)**.

[2] Minsung Kim, **Dong-Kyum Kim**, Jea Kwon, Nakyeong Yang, Kyomin Jung, and Meeyoung Cha. "How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models." *ACL* 2026.

[3] **Dong-Kyum Kim**, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. "Bilinear representation mitigates reversal curse and enables consistent model editing." *ICLR* 2026.

[4] Nakyeong Yang, **Dong-Kyum Kim**, Jea Kwon, Minsung Kim, Kyomin Jung, and Meeyoung Cha. "Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning." *ICLR* 2026.

[5] Jea Kwon, Jiwon Kim, **Dong-Kyum Kim**, and Meeyoung Cha. "Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing." *ICML 2026 Workshop on Mechanistic Interpretability.*

[6] Minsung Kim, Jea Kwon, **Dong-Kyum Kim**, and Meeyoung Cha. "In Search of the Engram in LLMs." *ICLR 2025 Blogpost.*

[7] Gwangsu Kim, **Dong-Kyum Kim**, and Hawoong Jeong. "Spontaneous emergence of rudimentary music detectors in deep neural networks." *Nature Communications* 2024.

[8] Jea Kwon, Sunpil Kim, **Dong-Kyum Kim**, Jinhyeong Joo, SoHyung Kim, Meeyoung Cha, and C. Justin Lee. "SUBTLE: An unsupervised platform with temporal link embedding that maps animal behavior." *IJCV* (2024).

[9] **Dong-Kyum Kim**†, Jea Kwon†, Meeyoung Cha, and C. Justin Lee. "Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity." *NeurIPS* 2023.

[10] Sangyun Lee, **Dong-Kyum Kim**, Jong-Min Park, Won Kyu Kim, Hyunggyu Park, and Jae Sung Lee. "Multidimensional entropic bound." *Phys. Rev. Research* **5**, 013194 (2023).

[11] **Dong-Kyum Kim**†, Sangyun Lee†, and Hawoong Jeong. "Estimating entropy production with odd-parity state variables via ML." *Phys. Rev. Research* **4**, 023051 (2022).

[12] Youngkyoung Bae, **Dong-Kyum Kim**, and Hawoong Jeong. "Inferring dissipation maps from videos using CNNs." *Phys. Rev. Research* **4**, 033094 (2022).

[13] **Dong-Kyum Kim** and Hawoong Jeong. "Deep reinforcement learning for feedback control in a collective flashing ratchet." *Phys. Rev. Research* **3**, L022002 (2021).

[14] **Dong-Kyum Kim**†, Youngkyoung Bae†, Sangyun Lee, and Hawoong Jeong. "Learning Entropy Production via Neural Networks." *Phys. Rev. Lett.* **125**, 140604 (2020).

[15] **Dong-Kyum Kim**†, Byunghwee Lee†, Daniel Kim, and Hawoong Jeong. "Multi-label classification of historical documents using hierarchical attention." *J. Korean Phys. Soc.* **76**, 368 (2020).

Technical Skills

- **Daily driver**: Currently running multiple Claude Code sessions (Max plan, $200/mo) via tmux as my primary research workflow.
- **ML/Interp**: PyTorch, JAX, TransformerLens, & nnsight.
- **Infrastructure**: Distributed training & inference (Slurm system with hundreds of H100s), large-scale data pipelines

Academic Service

Program Committee: AAAI 2027, CIKM 2026.
Reviewer: TMLR, NeurIPS 2026, ICML 2026 Workshop on Mechanistic Interpretability, ICML 2026 (Received Gold Reviewer Award), ICLR 2026, NeurIPS 2025, ICML 2025 Workshop World Models, ICLR 2025 Workshop Re-Align, NeurIPS 2024 Workshop Behavioral ML, ICML 2024 Workshop LLMs and Cognition.

Awards

Pre-doctoral Fellow of Physics, KAIST 2021

In the Press

- **Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity** (*NeurIPS* 2023): IBS Research News (English), IBS Research News (Korean), Donga Science, and EurekAlert!.
- **Learning Entropy Production via Neural Networks** (*Phys. Rev. Lett.* **125**, 140604, 2020): Physics and High Technology.

References

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